

Dental Image Data Generation for Instance Segmentation using Generative Adversarial Networks

Seok-Mo Heo^{4,5,†}, SeongJun Jung^{3,†}, HyeonMin Kwak^{1,2}, YongHo Jeong²,
SeongMin Yang^{1,2}, YongHak Lee^{1,2}, SungHwan Kim^{1,2,*}

¹Department of Applied Statistics, Konkuk University, Seoul 05029, Republic of Korea

²AI Analytics Team, Mustree, Seoul 05029, Republic of Korea

³IPAI, Seoul National University, Seoul 08826, Republic of Korea

⁴Department of Periodontology, School of Dentistry, Jeonbuk National University, Jeonju 54896, Republic of Korea

⁵Research Institute of Clinical Medicine of Jeonbuk National University, Jeonju 54907, Republic of Korea

(Received October 5, 2023; Revised November 9, 2023; Accepted November 13, 2023)

ABSTRACT

Deep learning techniques have begun applied to the practice of medicine. Deep learning may, for example, assist physicians by showing pathogenic structures within diseases, including cysts, tumors, or infected tissue, or by recognizing objects such as cellular nuclei, or tumorous liver growths. However, the development of diagnostic deep learning techniques has been hampered by a scarcity of clinical image data. This is particularly true in the dental field. Accordingly, we propose a method of generating dental images using Generative Adversarial Networks (GANs). In this paper, we compare the performance of a Mask R-CNN model for image segmentation trained on a dataset that included GAN-generated data against a Mask R-CNN model trained with a dataset that did not include GAN-generated data. Ultimately, we observed improved performance by the Mask R-CNN that employed our GAN-generated images.

Key words : Instance segmentation, Generative adversarial networks, Deep learning for dental images

1. Introduction

In recent years, remarkable advances in deep learning technology have attracted increasing attention from researchers and practitioners in fields as diverse as education, sociology, eco-

nomics, and anthropology [1]. With the development of deep neural networks, the ability to process ‘big data’ has increased and available computing power (e.g. GPU) has exploded [2]. As a result, today’s state-of-the-art approaches demand the application of deep learning techniques. For medical practitioners, these developments will have a significant impact on image processing [3-5] data analysis, diagnostics [6,7] and the provision of healthcare services.

Convolutional Neural Networks (CNNs) have achieved tremendous success in visual tasks such as image classification, object recognition, and image segmentation as many studies have been conducted in recent years. With the development of deep learning such as CNNs, many achievements have also been made in the medical field [8]. However, the lack of medical data is hindering the development of these innovative tech-

[†]These two authors contributed equally to this work as co-first authors.

* Correspondence should be addressed to Dr. SungHwan Kim, Associate Professor, Department of Applied Statistics, Konkuk University, Seoul 05029, Republic of Korea. Tel: +82-2-450-3658, Fax: +82-0504-376-3693, E-mail: shkim1213@konkuk.ac.kr

Seok-Mo Heo is a professor, Department of Periodontology, School of Dentistry, Jeonbuk National University, Jeonju 54896, Republic of Korea.

SeongJun Jung is a graduate student, IPAI, Seoul National University, Seoul 08826, Republic of Korea.

YongHo Jeong is a Ph.D. in physics, AI analytics Team, Mustree, Seoul 05029, Republic of Korea.

HyeonMin Kwak, SeongMin Yang, and YongHak Lee are a graduate students, Department of Applied Statistics, Konkuk University, Seoul 05029, Republic of Korea.

nologies. Basic medical knowledge is a necessary prerequisite for the refinement and annotation of medical data, and this makes it costly to gather such data. This problem of data shortage is the cause of over-fitting or poor performance. Therefore, medical domains need research to solve problems in limited environments such as lack of data.

A widely known way to solve this data shortage problem is to use data augmentation techniques. Typical data augmentation techniques include kernel filtering, geometric transformation, randomization, image mixing and color space transformation. It is known that this method can solve over-fitting problems or low performance problems [9,10]. Yadav et al. [11] experimentally demonstrated that data augmentation techniques such as data rotation and enlargement improve performance in medical image data-based disease diagnosis problems. However, Zhao et al. [12] argues that typical manual tuning approaches for data augmentation fail to capture complex variations. In particular, in the tooth segmentation problem covered in our study, the tooth image data varies greatly from person to person and from oral disease to person to person. Therefore, in order to overcome the data shortage problem by reflecting this problem, it is necessary to create a new high-quality image rather than simply transforming the image data. This is because it can have a similar effect to obtaining a tooth image of someone different from the existing tooth image or someone suffering from a different oral disease. Therefore, we show lack of predictive performance due to lack of data in tooth segmentation problems and show sophisticated results by applying data augmentation techniques that generate new, high-quality images rather than data augmentation techniques based on simple data deformation.

Methods of altering existing images that utilize deep learning include Adversarial Training, Neural Style Training, and Generative Adversarial Networks (GANs) [13]. The generative model allows us to create a new dataset while preserving the similar attributes as the original dataset. Due to the dramatically improved performance of GAN, there has been a growing interest in how to apply GAN to generate data. Through these networks, the generated data can be newly trained to create classification models that perform better. We used two approaches to generate an expanded set of images. Dental images were first generated using a GAN. Since the work of generating complex images in a GAN is prone to failure, we pre-trained the GAN's Generator (G) and Discriminator (D) to generate better images. We collected dental image data of anterior teeth with the permission of the Institutional Review Board of Jeonbuk National University Hospital. Using a dental noise function, the random

erasing method was applied to each teeth image. Then, to show how the generalization performance is, we generated the same number of simulation data and conducted an experiment like after applying the same method. We then applied the noise function to raw teeth images to create poor-quality teeth images. We trained the GAN using the generated parallel data. The GAN was applied to the original teeth data, resulting in 734 teeth images (i.e., the original 317 teeth images and 317 images created by the noise function). We used Mask R-CNN [14,15] for instance segmentation to prove the effectiveness of our method. Ultimately, our method resulted in improvements across numerous metrics; according to Table 2, we achieved 0.67% of Bounding Box IoU Average Recall (AR) and 0.75% of Segmentation IoU Average Precision (AP) during our simulation test. Since false negative is a significant metric in the medical field, the improvement in Recall illustrates how GAN-based data generation results in better-trained deep neural networks. The remainder of this paper is organized as follows. In Section 2 we provide a background to our study. Section 3 describes our methods. Section 4 explains how we set up our experiment and our experimental results are provided and discussed in Section 5.

2. Related Work

2.1. Data generation

Data generation is a way of creating new data while preserving the characteristics of the training data. Where data are very scarce, such generated data are very useful for training models, often as helpful as obtaining additional labeled data. According to [16], data generation was developed as a means of building powerful and useful deep learning models and reducing over-fitting from training on an original dataset. There are numerous ways to augment a limited dataset. For images, common methods include resizing, cropping, rotation, and translation.

2.2. Data augmentation based on data transformation

Methods such as rotation, zooming, horizontal flip, shear transformation and mix-up are generally used to improve the generalization ability of deep learning models. Rotation, zooming, horizontal flip and shear transformation are methods of augmenting data by changing the coordinates of data, and mix-up is a method of augmenting data through weighting-combina-

tion of image pairs from training data. Takahashi et al. [17] proposed Random Image Crop and Matching (RICAP) and tested its efficiency along with traditional data augmentation methodologies. In tasks such as Urinary Stones Segmentation in Abdominal X-Ray Images, the number of abdominal x-ray images containing urinary stones(s) is very small compared to normal abdominal x-ray images. Therefore, Preedanana et al. [18] proposed stone embedding augmentation, a medical image augmentation technique suitable for realistic appearance, showing better performance than the existing method in medical image segmentation.

2.3. Data augmentation based on deep learning

Generative Adversarial Networks (GANs) is a new strategy for data generation. GAN is adequate to over-sampling minority classes because it has the advantage of being able to generate synthesized images in an end-to-end layer fashion without any rules or constraints in an imbalanced dataset. Tanaka et al. [19] generated synthetic data for cancer detection problem. And they proved that the classifier performed better when it was trained on augmented data rather than only on the original dataset. Obtaining sufficiently large quantities of annotated medical images is very challenging. By applying GAN to data generation, Yi, Xin et al. [20] presents there are many researches done and these models result in better performance in biomedical image analysis. According to [20], using models trained on GAN generated data results in superior image analysis. GAN first proposed by Ian Goodfellow [21], learns generate new data using the train data set. A GAN consists of two networks [22]: Generator (G) and Discriminator (D). In typical GAN, G takes a random noise vector (z) and accordingly generates a fake image, ($G(z)$). D then receives an input image and decides if it is a true image (y) or synthesized image from generator ($G(z)$). The objective function of a vanilla GAN can be formulated as:

$$\min_G \max_D V(D, G) \\ = E_{x \sim p_{\text{data}}(x)} [\log D(x)] + E_{z \sim p_z(z)} [\log(1 - D(G(z)))] \quad (1)$$

where the G operates to minimize the objective function while the D tries to maximize it inversely. GAN have garnered significant attention from researchers, and have been vigorously studied for their potential applications across a diverse array of fields. Wasserstein GAN (WGAN) [23] solves the mode collapse and the difficulty of determining the model performance which are original GAN's problem by applying EM distance to

GAN, Super-Resolution GAN (SRGAN) [24] is used to increase the image resolution, and Speech Enhancement GAN (SEGAN) [25] filters out noise in recorded speech. Image-to-image translation using GAN models are focused on a research topic fervently. Cycle-consistent adversarial networks (CycleGAN) [26] and Discover Cross-Domain Relations with Generative Adversarial Networks (DiscoGAN) [27] enable image-to-image translation even if two or more domains of the image dataset are unmatched. GAN's loss function is expressed as:

$$LGAN(G, D) \\ = E_{(x,y)} [\log(D(x, y))] + E_{(x,z)} [\log((1 - D(x, G(x, z))))] \quad (2)$$

where x is the input image, y is the real image that corresponds to x , and z is random vector noise.

2.4. Instance segmentation

The Mask R-CNN [14] is a model for instance segmentation and an updated version of previous computer vision models (i.e. Fast R-CNN [28], Faster R-CNN [29]). Mask R-CNN consists of three stages, as shown in Fig. 1. In the first stage, Mask R-CNN resizes the input images to 1024×1024 using bilinear interpolation and padding for application to a backbone network. Next, Mask R-CNN passes input images into a backbone network (e.g. ResNet-101) and a Feature Pyramid Network (FPN) extracts the feature maps. which offers improved accuracy and computing speed. The Region Proposal Network (RPN) then detects the Regions of Interest (ROI). Category prediction and bounding box prediction are then implemented in parallel with mask prediction. In the final stage, the objective function of the Mask R-CNN model, which consists of the classification loss function, the bounding box loss function, and the mask loss function, is calculated by the softmax, the box offset regressor, and the mask FCN predictor, respectively.

3. Proposed Method

In this paper, we propose a method for GAN-based data generation, as shown in Fig. 2. We then apply this method to improve the performance of a Mask R-CNN model of image instance segmentation. Our purpose is to solve the data scarcity problem for dental images by generating artificial dental images. Using GAN, we achieved our goal. We also achieved improved performance of a teeth instance segmentation model using our data generation method.

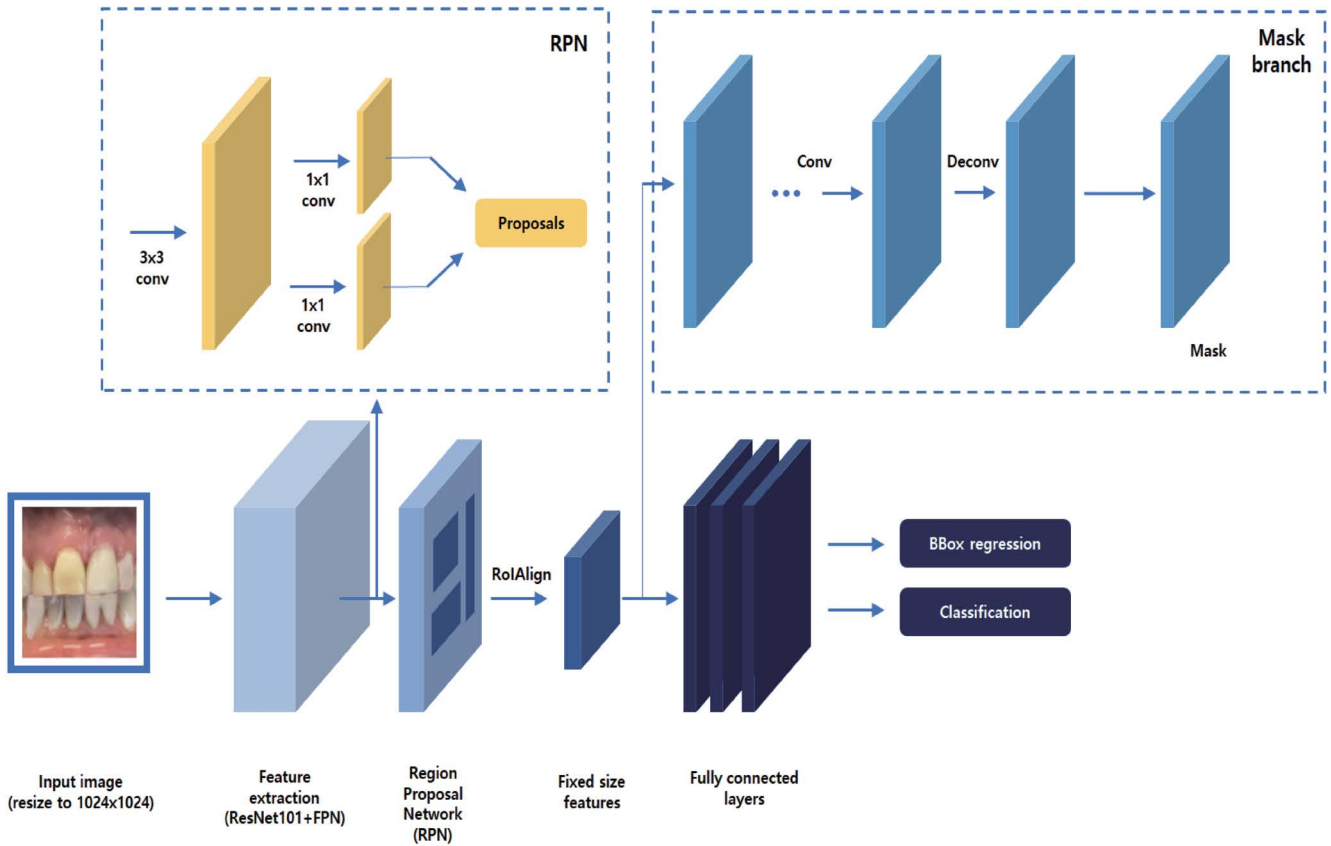


Fig. 1. The Overall architecture of Mask R-CNN.

3.1. Preprocessing

We separated 317 images of teeth into images of healthy and unhealthy teeth. Applying the noise function to all teeth images, poor-quality versions of the images were generated. In some of these images, teeth were removed from the images. At the end of this process, we obtained 317 poor-quality versions of teeth images synthesized, a parallel version of our original teeth dataset.

3.2. Training strategy

GAN are generative models that learn mapping from input images (z) to generate output images (y), ($G: z \rightarrow y$). G is trained to produce images that are not rejected as a negative sample by the adversarial D . GAN's discriminator D is a classifier that distinguishes between actual and the data generated by the Generator G . Our method is an adaptation of the G and D architectures model proposed by [30]. Using this GAN, we generated images of teeth in good condition based on images of teeth in poor condition, as shown in Fig. 3.

The GAN was fitted using the parallel teeth data generated

by the process previously discussed. However, it is known that GAN is usually not easy to train. One of the reasons why GAN is difficult to train is that if the generator model and the discriminator model are trained at the same time, the improvement of one model can lead to the sacrifice of another model. Therefore, we each used a method of training individually. First, we pre-trained the generator with the parallel dataset. We used a U-net [31] based generator. ResNet34 [32] was used for down-sampling U-net architecture. We used this generator to generate artificial teeth images with higher resolution. The generator was then applied to all raw teeth images to generate 317 fake images. We pre-trained the D using these generated images, after which we combined the two pre-trained models into the GAN to train the parallel datasets. The trained GAN was then used to generate images of teeth in good condition using images of teeth in poor condition. In this way, GAN generated 120 teeth images and the noise function generated 317 teeth images. Ultimately, we were able to obtain data for 754 teeth images, including raw data, GAN generation data, and noise function generated data.

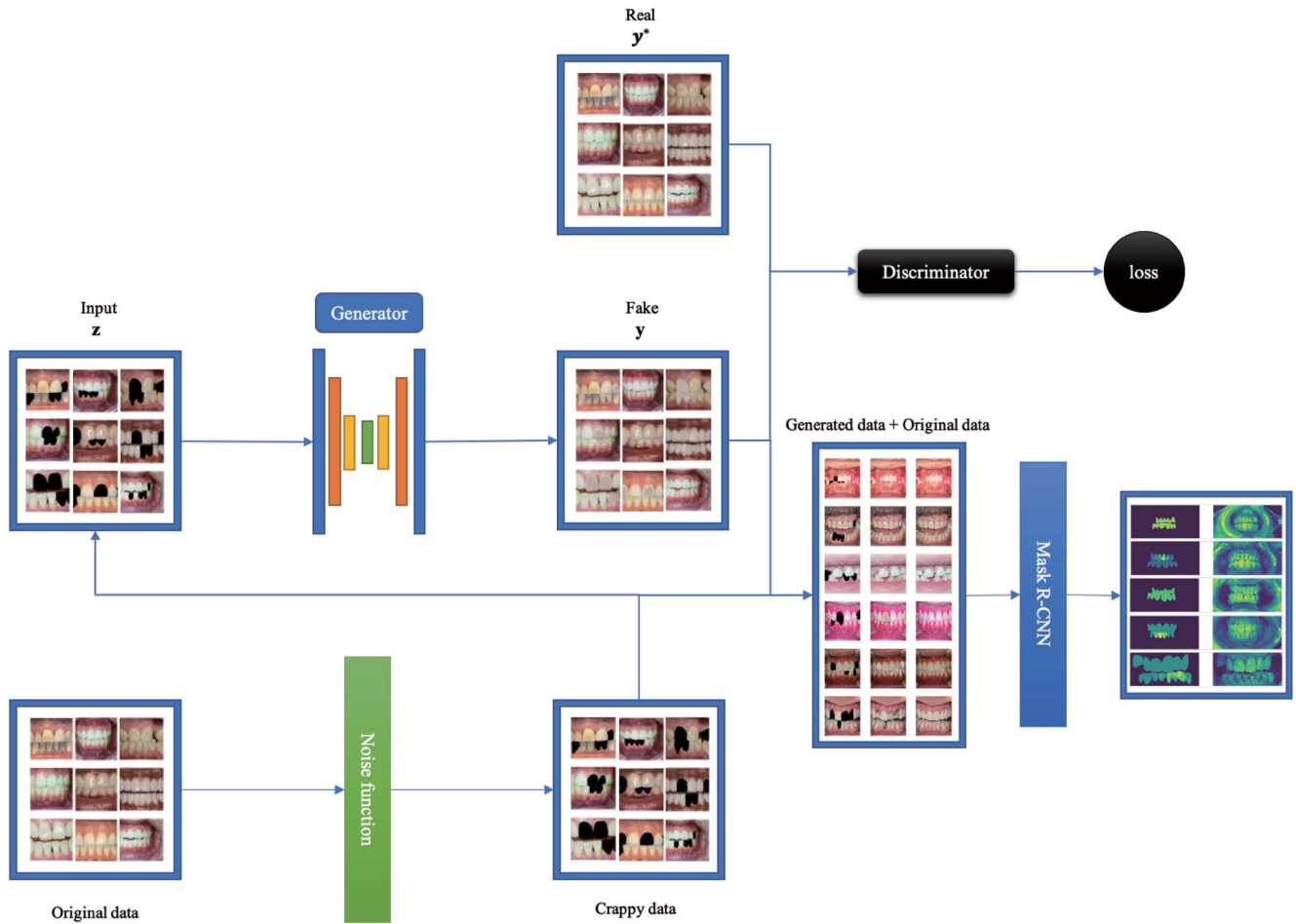


Fig. 2. The Conditional Generative Adversarial Network (CGAN) model architecture.

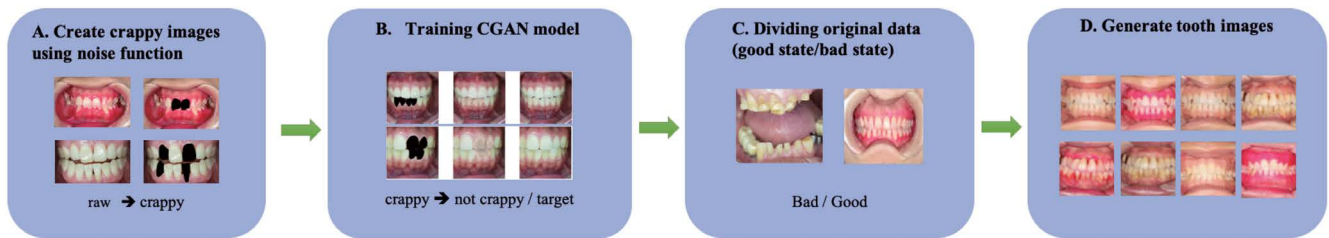


Fig. 3. Overview of Data generation using Generative Adversarial Networks: (A) creation of manipulated images using noise function, (B) GAN training with generated parallel teeth image (manipulated $\rightarrow$ not manipulated), (C) dividing raw teeth images into good condition and bad condition, (D) creating teeth data by applying GAN to teeth of bad condition.

4. Experimentation

To show the effect of the data generation, we compared the performance of a Mask R-CNN model trained on the larger dataset that included newly created images against the same model trained on a dataset that did not include generated data. We then generated teeth simulation data to prove that it works

for general data as well as specific data. With our method, we improved the performance of the Mask R-CNN.

4.1. Implementation details

In the training process, each image is resized first to 1024×1024 by Bi-linear interpolation. GAN's Generator model U-net

for image generation was initialized with trained weights based on image-net data and trained with our data. In addition, Adam optimizer was used for our network training and hyper parameters were set to default values (initial learning rate $\gamma=0.001$, $\beta_1, \beta_2=(0.9, 0.999)$, $\varepsilon=1e-08$, $\lambda=0$). Our network is implemented based on pytorch 1.10.1 + cu113 [33]. Both training and testing are conducted on an 16-core, 32 threads PC with an AMD Ryzen-9 5950X CPU (128 GB RAM) and a RTX 3080ti GPU (12 GB memory).

4.2. Preprocessing

4.2.1. Simulation data

We used the python PIL library to generate 317 simulated teeth images to be used for training and testing. The simulated data was designed such that 12 differently-sized rectangular teeth shapes were in the center of the image, with gum shapes appearing around the teeth. Teeth positions were randomly chosen based on a uniform distribution. All images had a resolution of $512 \times 512 \times 3$ pixels.

4.2.2. Real dental image data

We obtained 317 teeth images from Jeonbuk National University Dental Hospital pursuant to IRB exemption approval (No: 2020-06-007). Using these images, we annotated 12 anterior teeth (i.e. six upper teeth and six lower teeth). To apply the GAN, each teeth image was classified into a ‘good condition’

and a ‘poor condition’ on the basis of whether all 12 teeth appeared healthy.

4.3. Data generation

We applied the noise function to our 317 raw teeth images to make 317 poor-quality teeth images. We can obtain 754 teeth images by applying GAN and noise function. The GAN’s Generator model U-net was pre-trained on this generated parallel image set. GAN’s D was pre-trained after the G was applied to 317 poor-quality teeth images to generate fake teeth images. The discriminator is a classification model composed of Convolution, Batchnorm, ReLU, and Residual modules that distinguishes fake from authentic images. The pre-trained G and D were combined to form the GAN. This combined GAN model was then retrained on the parallel data generated during the process described above, and then applied to the original images showing teeth in poor condition and used to generate 120 images of teeth in good condition. Thus, we ultimately obtained 754 teeth images by combining the original dataset of 317 teeth with the data of 120 teeth created by GAN and 317 noise function.

4.4. Enhancing mask R-CNN performance

To illustrate the effects of data generation, we trained a Mask R-CNN for teeth instance segmentation using two different

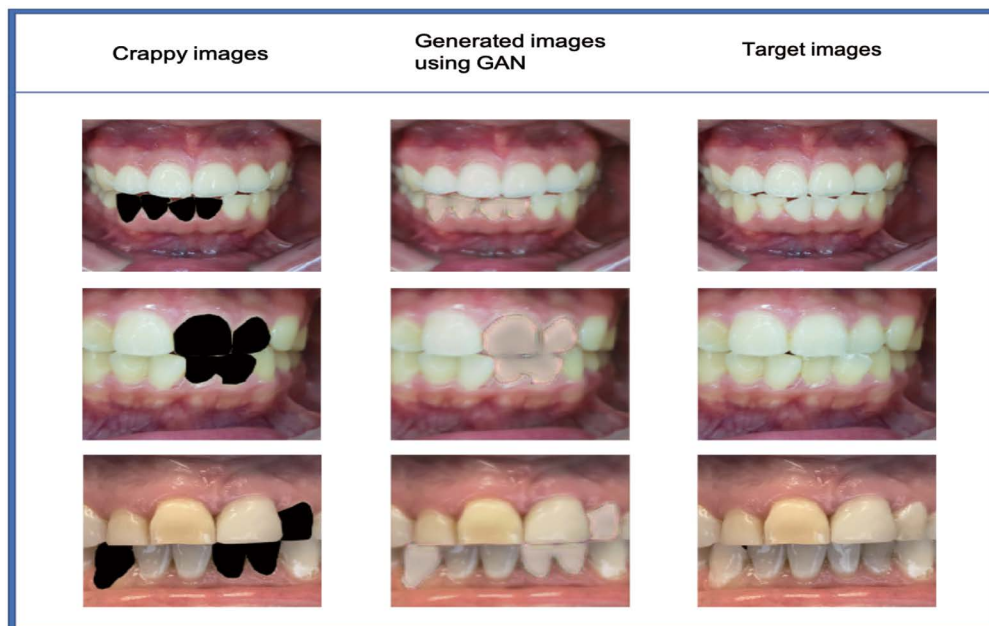


Fig. 4. Example results of Generative Adversarial Networks on real teeth data.

datasets - one with and one without the generated data - and compared the accuracy of the models. The Mask R-CNN's network architecture largely consists of two components: (1) feature extraction and (2) instance segmentation. A ResNet101 module, pre-trained with the COCO dataset, was used as a backbone network. This was used alongside a feature pyramid network (FPN) architecture, which is designed to extract features, to achieve greater accuracy and processing speed. The model then detects ROI in the head of the network architecture, and from this detection and classification are performed. By these processes a fully convolutional mask prediction is implemented for instance segmentation. The Fig. 4 shows the results of the Mask R-CNN model with and without generated data. As Fig. 4 makes clear, the model that applied generated data expressed the teeth boundary in more detail and detected teeth more accurately.

4.5. Results

4.5.1. Evaluation metric

We used Average Precision (AP) and Average Recall (AR) as evaluation metrics. Precision refers to the ratio of the number

of samples that actually belong to the positive class to the number of samples that are the output to belong to the training class. Recall refers to the ratio of the number of samples output as belong to the positive class to the number of samples belong to the positive class. Higher values indicate improved predictive power. Intersection over Union (IoU), also known as the Jaccard index or the Jaccard similarity coefficient, is a means of gauging the similarity and diversity of sample sets. An algorithm that provides a predicted bounding box as output can be evaluated using IoU. To evaluate object detectors using IoU, we must manually label the ground true bounding box and the predicted bounding boxes from our model.

$$\text{Precision} = \frac{\text{TruePositive}}{\text{TruePositive} + \text{FalsePositive}} \quad (3)$$

$$\text{Recall} = \frac{\text{TruePositive}}{\text{TruePositive} + \text{FalseNegative}} \quad (4)$$

We calculated these metrics using the Microsoft COCO evaluation code. AP and AR were averaged over multiple Intersection over Union (IoU) values. Specifically, we used 10 IoU thresholds from 0.5 to 0.95 at 0.05 intervals; AP and AR are the

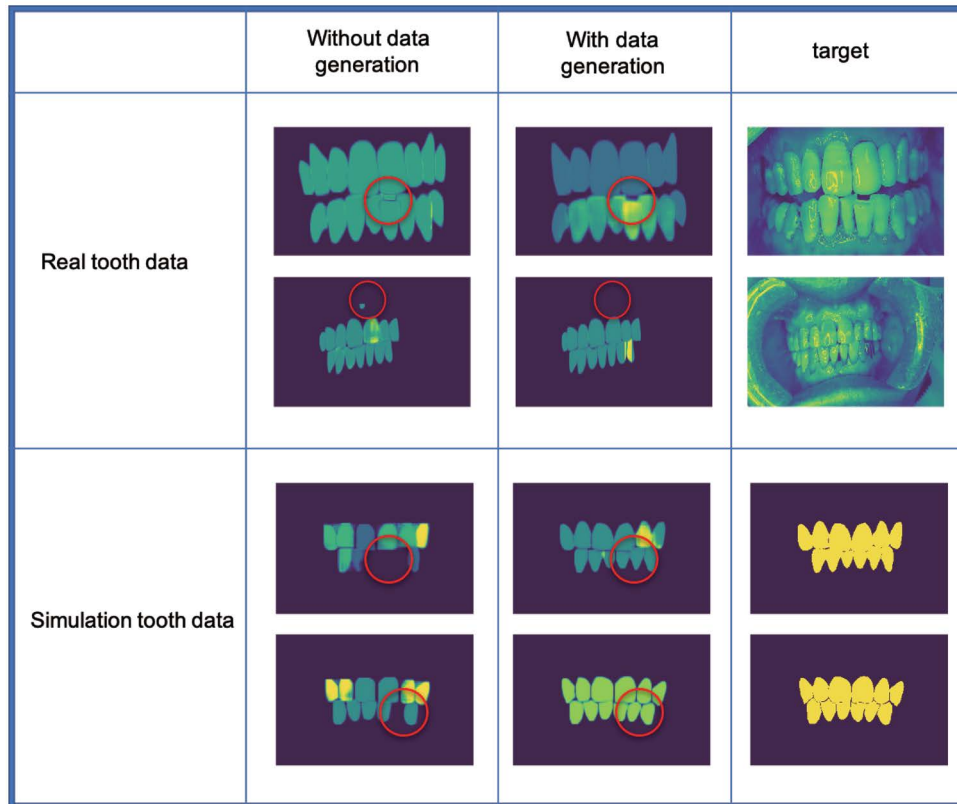


Fig. 5. Prediction results of the Mask R-CNN model with data generation applied and the prediction results of the Mask R-CNN model without data generation.

average accuracy of the predictions that are above the minimum threshold of IoU. AR is the maximum recall when the number of detections for each image is fixed, and is averaged across categories and IoUs. The detection evaluation metrics using bounding boxes and segmentation masks are the same in all respects except for the IoU calculations (performed for each box or mask).

$$J(G, P) = \frac{|G \cap P|}{|G \cup P|} = \frac{|G \cap P|}{|G| + |P| - |G \cap P|} \quad (5)$$

4.5.2. Comparative method

We compared AP and AR in models that were trained with generated data against models that were not. For all AP, the average value applied by IoU threshold was from 0.5 to 0.95 at

0.05 intervals and the applied AP by IoU threshold of 0.5 and 0.75 were calculated. The max detections per image of AP was set to 100. For all AR, the average value applied by IoU threshold was from 0.5 to 0.95 at 0.05 intervals. AR max detections per image were set at either 1, 10 or 100. The results of using real and generated teeth data were compared.

4.5.3. Comparison results

Looking at Table 1, data generation applied to real teeth data did not increase all metrics. However, all but metrics have increased. In particular, the Bounding box AP(IoU=0.5: 0.95) increased from 0.167 to 0.18. In addition, Segmentation AP (IoU=0.5: 0.95) also increased from 0.156 to 0.172. The results of the experiment can be used to prove that our data generation method is effective.

Table 1. Average precision and average recall without simulation teeth data

IOU metric		Augmented	Raw
Bounding box	Average Precision (IoU = 0.50:0.95, area = all, maxDets = 100)	0.180	0.167
	Average Precision (IoU = 0.50, area = all, maxDets = 100)	0.281	0.273
	Average Precision (IoU = 0.75, area = all, maxDets = 100)	0.173	0.152
	Average Recall (IoU = 0.50:0.95, area = all, maxDets = 1)	0.035	0.035
	Average Recall (IoU = 0.50:0.95, area = all, maxDets = 10)	0.300	0.301
	Average Recall (IoU = 0.50:0.95, area = all, maxDets = 100)	0.351	0.352
Segmentation	Average Precision (IoU = 0.50:0.95, area = all, maxDets = 100)	0.172	0.156
	Average Precision (IoU = 0.50, area = all, maxDets = 100)	0.237	0.225
	Average Precision (IoU = 0.75, area = all, maxDets = 100)	0.173	0.152
	Average Recall (IoU = 0.50:0.95, area = all, maxDets = 1)	0.035	0.034
	Average Recall (IoU = 0.50:0.95, area = all, maxDets = 10)	0.296	0.297
	Average Recall (IoU = 0.50:0.95, area = all, maxDets = 100)	0.345	0.347

Table 2. Average precision and average recall with simulation teeth data

IOU metric		Augmented	Raw
Bounding box	Average Precision (IoU = 0.50:0.95, area = all, maxDets = 100)	0.670	0.330
	Average Precision (IoU = 0.50, area = all, maxDets = 100)	0.881	0.594
	Average Precision (IoU = 0.75, area = all, maxDets = 100)	0.794	0.400
	Average Recall (IoU = 0.50:0.95, area = all, maxDets = 1)	0.072	0.051
	Average Recall (IoU = 0.50:0.95, area = all, maxDets = 10)	0.631	0.383
	Average Recall (IoU = 0.50:0.95, area = all, maxDets = 100)	0.734	0.428
Segmentation	Average Precision (IoU = 0.50:0.95, area = all, maxDets = 100)	0.750	0.388
	Average Precision (IoU = 0.50, area = all, maxDets = 100)	0.882	0.582
	Average Precision (IoU = 0.75, area = all, maxDets = 100)	0.809	0.468
	Average Recall (IoU = 0.50:0.95, area = all, maxDets = 1)	0.078	0.061
	Average Recall (IoU = 0.50:0.95, area = all, maxDets = 10)	0.697	0.435
	Average Recall (IoU = 0.50:0.95, area = all, maxDets = 100)	0.813	0.483

Table 3. Model performance comparison table according to the amount of data generation

IOU metric		400	450	500
		Augmented	Augmented	Augmented
Bounding box	Average Precision (IoU = 0.50:0.95, area = all, maxDets = 100)	0.608	0.585	0.684
	Average Precision (IoU = 0.50, area = all, maxDets = 100)	0.833	0.891	0.922
	Average Precision (IoU = 0.75, area = all, maxDets = 100)	0.733	0.757	0.822
	Average Recall (IoU = 0.50:0.95, area = all, maxDets = 1)	0.067	0.063	0.073
	Average Recall (IoU = 0.50:0.95, area = all, maxDets = 10)	0.597	0.562	0.627
	Average Recall (IoU = 0.50:0.95, area = all, maxDets = 100)	0.672	0.644	0.732
Segmentation	Average Precision (IoU = 0.50:0.95, area = all, maxDets = 100)	0.699	0.716	0.785
	Average Precision (IoU = 0.50, area = all, maxDets = 100)	0.834	0.893	0.919
	Average Precision (IoU = 0.75, area = all, maxDets = 100)	0.778	0.818	0.860
	Average Recall (IoU = 0.50:0.95, area = all, maxDets = 1)	0.073	0.072	0.079
	Average Recall (IoU = 0.50:0.95, area = all, maxDets = 10)	0.666	0.666	0.707
	Average Recall (IoU = 0.50:0.95, area = all, maxDets = 100)	0.751	0.769	0.832

In Table 2, we compared the performance of Mask R-CNN with and without data generation applied to simulation data. The data generation applied to the simulation data increased the metric for all conditions. Especially in the segmentation metric part, a significant increase in the metrics can be confirmed. By showing that data generation is effective in simulation data, the generalized performance of the method in this paper can be proved.

We experimented with the model's performance change depending on the amount of data generated. We were able to obtain results that improved the performance of the model based on increasing the amount of data generated. In Table 3, most metrics can be seen to improve based on increasing the amount of data generated.

5. Conclusions

Successful deep learning requires vast quantities of data. However, to date, most deep learning tools in medicine have relied on relatively small datasets to perform the tasks demanded of them. The ability to increase the size of a training dataset benefits not only medicine but also other fields that are affected by potential applications of deep learning. In this work, we proposed a GAN-based data generation solution to enhance the performance of instance segmentation of dental data. Using Pix2Pix, we produced synthetic augmented images for our teeth and simulation data to provide additional information based on images from training dataset. Using GAN, we produced syn-

thetic augmented images to be added to our training dataset. Where an adequate amount of suitable data is difficult to obtain, data generation is an inevitable fallback method for training deep learning models since model over-fitting is reduced by data generation. We have shown that by using GAN-based methods, false predictions can be sufficiently reduced and sophisticated predictions can be made. However, there can be many unexpected cases in medical images, and it can be hard to say that predictions of unseen data are also made precisely. Therefore, there is a need for a way to predict sufficient performance and a generation method to create various cases even in domains that have not been seen. And it is known that GAN-based data augmentation methods are often unstable in learning. Therefore, in this study, a relatively simple GAN-based ADAM optimizer was used for a simple method, but in future studies, I think the score-based generative model and optimization method should also be studied. We did not consider traditional data generation methods in this paper, but Bowles, Christopher et al. [34] notes that using both traditional and GAN-based data generation together is superior to using either method alone. In subsequent research, we will consider how other GAN models, like those introduced by [35] and [36], could be used to further expand training datasets.

Acknowledgements

This paper was supported by Konkuk University in 2022.

References

1. Min JH, Jung SJ, Jung SH, Yang SM, Cho JS, Kim SH. Grammatical error correction models for Korean language via pre-trained denoising. *QBS* 2020;39:17-24.
2. Jung SH, Yang SM, Lee ES, Lee YH, Ko JS, Lee SJ, et al. Estimation of particulate levels using deep dehazing network and temporal prior. *J Sens* 2020;2020:1-9.
3. McAuliffe MJ, Lalonde FM, McGarry D, Gandler W, Csaky K, Trus BL. Medical image processing, analysis and visualization in clinical research. *Proceedings 14th IEEE Symposium on Computer-Based Medical Systems. CBMS 2001. IEEE. 2001. p. 381-386.*
4. Razzak MI, Naz S, Zaib A. Deep learning for medical image processing: Overview, challenges and the future. *Classification in BioApps. Springer. 2018. p. 323-350.*
5. Ko KI, Gwak HM, Lee ES, Kim GH, Moon DH, Cho YJ, et al. Action recognition model to monitor illegal dumping using zoom-in image sequence data. *QBS* 2021;40:95-102.
6. Agravat RR, Raval MS. Deep learning for automated brain tumor segmentation in MRI images. *Soft Computing Based Medical Image Analysis. Elsevier. 2018. p. 183-201.*
7. Park YE, Kim SH, Chon KM, Lee HK, Kim JH, Shin JK. Impacts of heavy rain and floodwater on floating debris entering an artificial lake (Daecheong Reservoir, Korea) during the summer. *Desalination Water Treat* 2021;219:399-404.
8. Kim SH, Heo SM, Yang SM, Kim YE, Han JS, Jung SH. Instance segmentation guided by weight map with application to tooth boundary detection. *QBS* 2020;39:159-167.
9. Han JS, Jo KC, Lim WT, Lee YH, Ko KM, Sim ES, et al. Reinforcement learning guided by double replay memory. *J Sens* 2021;2021:1-8.
10. Park YE, Lee HK, Shin JK, Chon KM, Kim SH, Cho KH, et al. A machine learning approach for early warning of cyanobacterial bloom outbreaks in a freshwater reservoir. *J Environ Manage* 2021;288:112415.
11. Yadav SS, Jadhav SM. Deep convolutional neural network based medical image classification for disease diagnosis. *J Big Data* 2019;6:1-18.
12. Zhao A, Balakrishnan G, Durand F, Guttag JV, Dalca AV. Data augmentation using learned transformations for one-shot medical image segmentation. *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition: 2019. p. 8543-8553.*
13. Lee H, Thoummala N, Park HJ, Ham SH, Yu JW, Hwang JH, et al. AI-based dental prostheses fabrication using generative adversarial networks. *QBS* 2021;40:39-44.
14. He K, Gkioxari G, Dollár P, Girshick R. Mask r-cnn. *Proceedings of the IEEE International Conference on Computer Vision: 2017. p. 2961-2969.*
15. Suh SM, Park YE, Ko KM, Yang SM, Ahn JH, Shin JK, et al. Weighted mask R-CNN for improving adjacent boundary segmentation. *J Sens* 2021;2021:1-8.
16. Shorten C, Khoshgoftaar TM. A survey on image data augmentation for deep learning. *J Big Data* 2019;6:60.
17. Takahashi R, Matsubara T, Uehara K. Rcap: Random image cropping and patching data augmentation for deep cnns. *Asian Conference on Machine Learning. PMLR. 2018. p. 86-98.*
18. Preedanana W, Suzuki K, Kondo T, Kobayashi M, Tanaka H, Ishioka J, et al. Urinary stones segmentation in abdominal X-ray images using cascaded U-Net pipeline with stone-embedding augmentation and session-size reweighting approach. *IEEE Access: 2023;11:25702-25712.*
19. Tanaka FHKdS and Aranha C. Data augmentation using GANs. *arXiv preprint arXiv: 1904.09135 2019.*
20. Yi X, Walia E, Babyn P. Generative adversarial network in medical imaging: A review. *Med Image Anal* 2019;58:101552.
21. Goodfellow I, Pouget-Abadie J, Mirza M, Xu B, Warde-Farley D, Ozair S, et al. Generative adversarial nets. *NIPS 27. Ed. by Ghahramani Z, Welling M, Cortes C, Lawrence ND, and Weinberger KQ. Curran Associates, Inc., 2014:2672-2680. Available from: <http://papers.nips.cc/paper/5423-generative-adversarial-nets.pdf>.*
22. Li Y, Shen L. cC-GAN: A robust transfer-learning framework for HEp-2 specimen image segmentation. *IEEE Access: 2018;6: 14048-14058.*
23. Arjovsky M, Chintala S, Bottou L. Wasserstein gan. *arXiv preprint arXiv:1701.07875 2017.*
24. Ledig C, Theis L, Huszár F, Caballero J, Cunningham A, Acosta A, et al. Photo-realistic single image super-resolution using a generative adversarial network. *Proceedings of the IEEE Conference on Computer Vision and Pattern Recognition: 2017. p. 4681-4690.*
25. Pascual S, Bonafonte A, Serra J. SEGAN: Speech enhancement generative adversarial network. *arXiv preprint arXiv:1703.09452 2017.*
26. Zhu JY, Park T, Isola P, Efros AA. Unpaired image-to-image translation using cycle-consistent adversarial networks. *Proceedings of the IEEE International Conference on Computer Vision: 2017. p. 2223-2232.*
27. Kim TS, Cha MS, Kim HS, Lee JK, Kim JW. Learning to discover cross-domain relations with generative adversarial networks. *arXiv preprint arXiv:1703.05192 2017.*
28. Girshick R. Fast r-cnn. *Proceedings of the IEEE International Conference on Computer Vision. 2015. p. 1440-1448.*
29. Ren S, He K, Girshick R, Sun J. Faster r-cnn: Towards real-time object detection with region proposal networks. *NIPS 2015. p. 91-99.*
30. Isola P, Zhu JY, Zhou T, Efros AA. Image-to-image translation with conditional adversarial networks. *Proceedings of the IEEE Conference on Computer Vision and Pattern Recognition. 2017. p. 1125-1134.*
31. Ronneberger O, Fischer P, Brox T. U-net: Convolutional networks for biomedical image segmentation. *International Conference on Medical image computing and computer-assisted inter-*

- vention. Springer. 2015. p. 234-241.
32. He K, Zhang X, Ren S, Sun J. Deep residual learning for image recognition. *Proceedings of the IEEE Conference on Computer Vision and Pattern Recognition*. 2016. p. 770-778.
33. Paszke A, Gross S, Massa F, Lerer A, Bradbury J, Chanan G, et al. Pytorch: An imperative style, high-performance deep learning library. *NIPS* 2019;32.
34. Bowles C, Chen L, Guerrero R, Bentley P, Gunn R, Hammers A, et al. Gan augmentation: Augmenting training data using generative adversarial networks. *arXiv preprint arXiv:1810.10863* 2018.
35. Yan W, Wang Y, Gu S, Huang L, Yan F, Xia L, et al. The domain shift problem of medical image segmentation and vendor-adaptation by Unet-GAN. *Medical Image Computing and Computer Assisted Intervention-MICCAI 2019: 22nd International Conference, Shenzhen, China, October 13-17, 2019, Proceedings, Part II* 22. Springer. 2019. p. 623-631.
36. Iqbal T, Ali H. Generative adversarial network for medical images (MI-GAN). *J Med Syst* 2018;42:1-11.